Construction

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This file constructs the data needed for modeling, then saves the result as allData.RData, so you would just need to execute Modelling.qmd

```
# Installing pacman
if (!require("pacman")) install.packages("pacman")
```

```
Loading required package: pacman
```

```
#Loading packages, install if necessary
pacman::p_load(readxl, quantmod)

#importing excel sheet
#Data: Robert Shiller, Yale university
#S&P Composite Stock Price Index, dividends, earnings
#http://www.econ.yale.edu/~shiller/data.htm
ie_data <- read_excel("data/ie_data.xls",
    sheet = "Data", range = "A8:P1872", col_types = c("text",
        "numeric", "numeric", "numeric",
        "numeric", "numeric", "numeric",
        "numeric", "numeric", "numeric",
        "numeric", "numeric", "text", "numeric",
        "text", "numeric"))
```

```
New names:
* `Price` -> `Price...8`
* `Price` -> `Price...10`
* `Earnings` -> `Earnings...11`
* `Earnings` -> `Earnings...12`
* `` -> `...14`
* `` -> `...16`
```

```r
# get wholesale price index from FRED
getSymbols("PPIACO", src = "FRED")
```

```
[1] "PPIACO"
```

```r
wpi <- PPIACO["1926-03-01/2026-03-01"]

ie_data$Date <- seq(from = as.Date("1871-01-01"),
                    to = as.Date("2026-04-01"),
                    by = "month")
df <- ie_data[ie_data$Date >= as.Date("1926-03-01") &
                ie_data$Date <= as.Date("2026-03-01"),]

price <- df$P
dividend <- df$D
log_price <- log(price)
log_dividend <- log(dividend)

save.image("allData.RData")
```